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Summary

A seasoned Senior Signalling Expert with over 16 years of extensive experience in delivering complex metro and rail systems, including CBTC and ETCS L2. Proven expertise in end-to-end project lifecycle management, from planning and design through testing, commissioning, and handover. Demonstrated ability to lead diverse teams, manage critical interfaces across civil and systems scopes, and ensure rigorous adherence to safety, quality, and international standards. Adept at stakeholder engagement and governance, with a strong track record of optimizing project delivery and achieving compliance in high-stakes environments. Possesses a deep understanding of V-Cycle systems engineering and risk management, positioning to effectively lead large-scale infrastructure projects.

Work Experience

Senior Signaling Engineer -T&C, Siemens Mobility SG,

11/2025 – Present

- Spearheaded Testing and Commissioning (T&C) activities for the CBTC system deployed on the DTL extension line.
- Executed Automatic Train Protection (ATP) and Automatic Train Operation (ATO) dynamic tests for the DTL extension area.
- Orchestrated test activities from the Depot Control Center.
- Compiled and submitted comprehensive test reports, meticulously logging any identified issues during testing.
- Analyzed test logs to perform troubleshooting and implement issue resolutions.
- Provided daily updates on test progress.

Head – Signaling and Telecom., Balaji Railroad Systems Pvt Ltd, BARSYL India,

05/2025 – 10/2025

- Directed key projects including Chennai Metro (Siemens/Hitachi CBTC), Tanzania ETCS L2 Upgradation, Saudi Land Bridge, and Etihad HSR.
- Optimized project management processes, achieving 100% compliance with RDSO and CENELEC standards.
- Defined technical specifications, budgets, vendor strategies, and cybersecurity protocols for telecommunication infrastructure.
- Championed digital transformation initiatives across signaling and telecommunication systems.

Independent Consultant – CBTC Design Review Expert, Independent Consultant – CBTC Design Review Expert,

01/2025 – 04/2025

- Provided expert technical guidance for project planning and Verification and Validation (V&V) activities, including the formulation of detailed plans and procedures.

- Validated system designs against technical specifications, operational requirements, and functional performance criteria.
- Reviewed Safety Integrity Level (SIL) documentation and supported safety certification processes.
- Verified the accurate design of interfaces between the CBTC system and associated subsystems: Rolling Stock, Interlocking Systems, Passenger Information Systems (PIS), Platform Screen Doors (PSD), and Operations Control Center (OCC).

Systems Integration Testing Engineering Lead., Queensland Rail, Brisbane,

10/2024 – 12/2025

- Directed system integration testing across multiple disciplines for the Cross River Rail project.
- Managed and allocated testing resources, ensuring adherence to project and international standards.

Senior Signaling Systems Validation Engineer., Thales/Hitachi KSA,

07/2022 – 09/2024

- Served as a Certified Trainer for ETCS L2 Signaling Subsystems.
- Delivered comprehensive Signaling training to Saudi Arabia Railway level 1 maintenance staff.
- Developed and finalized training manuals for maintenance personnel.
- Analyzed system and subsystem hardware and software requirements to define appropriate validation methodologies for the ETCS L2 THALES INTERLOCKING System.
- Verified system and subsystem level requirements through compatibility studies.
- Authored test procedures in compliance with system and subsystem hardware and software requirements.
- Developed compliance documents according to applicable standards and requirements specifications throughout the EN 50128 V&V lifecycle.
- Reviewed project lifecycles for adherence to standards.
- Reviewed and developed software system and subsystem level test procedures as needed.
- Monitored software testing activities to ensure compliance with standard requirements.
- Evaluated test reports and non-compliance findings, providing constructive feedback.
- Ensured designed software and hardware met all applicable requirements.
- Verified software and hardware test procedures.
- Developed and executed test procedures in accordance with applicable standards (e.g., EN 50128).
- Reviewed testing and commissioning progress and status.
- Performed troubleshooting, identified system and subsystem issues, proposed resolutions, and reported findings to engineering and management.
- Conducted a feasibility study for the upgradation of Dammam Marshalling Yard to Automatic Train Control Signaling.

Project Technical Lead – Signaling., Cyient India Pvt Ltd,

02/2021 – 05/2022

- Functioned as Project Technical Leader for the Etihad Rail project, holding full responsibility for reviewing Signaling Plan JB and DB placement allocation, and interfacing with the Hitachi offshore team.
- Oversaw Wayside Cable length calculations, Wayside Cable plans, and Wayside Wiring diagrams design review & Asset management.

- Served as the primary technical interface with Hitachi Customer for addressing technical queries.

ATC – RSK Interface Lead Engineer (Automatic Train Control vs Rolling Stock), Alstom Middle East,

03/2020 – 12/2020

- Served as the ATC System Lead Engineer, with complete accountability for the Automatic Train Control (ATC) system and its interfaces with all subsystems, including OCS (SCADA), Rolling Stock (TCMS), PSD, and FDS (Fire Distribution Systems).
- Supervised subcontractor progress on testing and commissioning activities for the ATC scope and its interfaces with various subsystems.
- Managed system testing for fire activation at different levels (Platform and Concourse), verifying the reaction of subsystems such as MEP (Smoke curtains, Lifts and Escalators, Emergency lights, Travellators, Sliding doors), IAS, CCTV, OCS, ECS, AFC, and ATC according to System Safety Assurance safety test cases.
- Participated in weekly EXPOLINK consortium meetings, reviewing progress and planning the upcoming four-week schedule with all subcontractors.
- Demonstrated a strong understanding of Safety Integrity Level (SIL) certifications.
- Provided technical review for all ATC test cases executed by the subcontractor.
- Exhibited thorough understanding and knowledge of EN 50126, EN 50128, and EN 50129 standards.
- Actively participated in HAZOP, FMEA, and Hazard Risk Assessment meetings, contributing insights related to ATC and its interfaces.
- Coordinated with the Independent Safety Assessor (ISA) on safety notice (SN) observations and comments, working towards issue resolution.
- Ensured ATC System readiness for the System Operation Demonstration Test (SODT).
- Managed the handover of trains to the customer for revenue operations.

ATC Testing and Commissioning Lead Engineer (Automatic Train Control - CBTC), Thales International Middle East,

04/2018 – 02/2020

- Functioned as the ATC Testing and Commissioning (T&C) Lead Engineer, with full responsibility for all safety-related test procedures and test cases.
- Developed safety test cases for the integration testing between the onboard Thales VOBC and Alstom Rolling Stock TCMS.
- Spearheaded the enhancement of the existing Dubai Metro Green and Red Lines, conducting static and dynamic testing of the Rashidiya Depot and Jebel Ali Depot, and overseeing the testing and commissioning of the R2020 extension Line.

Technical Writer (Telecom Security Sub Systems, Alstom Urbalis U400 CBTC), Thales Gulf,

09/2017 – 03/2018

- Reviewed the Requirement Traceability Matrix, identifying requirements pertinent to the Site T&C phase.
- Provided technical review for T&C procedures and maintenance manuals for various Telecom and Security subsystems, including FDS, CCTV, UPS, PDB, COM TV, SCADA, TETRA, and Telephony.
- Participated in technical review meetings with the client to discuss T&C procedures.
- Secured client approval for T&C documentation.
- Managed on-site Testing and Commissioning activities.

Independent Consultant – CBTC Design Review Expert, Independent Consultant – CBTC Design Review Expert,

03/2017 – 08/2017

- Provided expert technical guidance for project planning and Verification and Validation (V&V) activities, including the formulation of detailed plans and procedures.
- Reviewed the CBTC system architecture, encompassing onboard, wayside, and control center components.

FIVP (IVVQ) Validation Lead Engineer, Alstom Transport India Limited,

08/2015 – 02/2017

- Analyzed and deployed System Safety Rules in alignment with SIL4 Standards for project implementation.
- Managed Test Bench deployment for integration testing of all subsystems.
- Evaluated SYSTEM DATA BASE changes and updated Beacon layout & TLPM layout accordingly.
- Deployed ATS data & Software to test ATS functionalities, installing new versions of the ATS upon release.
- Managed ATC data deployment, including NVS & VS updates for ATC data or software.
- Authored T&C procedures for validating various ATC subsystems, such as ZC, IXL, ATS, CBI, CC (TRAIN), smart I/O cards, Trackside Equipment, SPKS, Beacons, and Point Machines.
- Developed test sheets and FIVP reports, ensuring successful FAT completion prior to system handover.
- Analyzed issues reported by the site T&C team, delegated issues to subsystem leaders, and logged change requests.

Systems Integration Test Engineer.| System Design Engineer.| Testing and Commissioning Eng., E to E Transportation Infrastructure Pvt Ltd,

11/2010 – 07/2015

- Engineered CBTC and Cut-in Design Circuits, including:
- Replacement of existing lamp signals to LED signals as part of Release 1 delivery.
- Design of Points Circuits, Track Circuits, COC Circuits, Staff Protection Key Switch (SPKS) Circuits, Emergency Stop Plunger (ESP) Circuits, Emergency Stop Button (ESB) Circuits, PMI I/O Interface, Platform Screen Door (PSD & HHPSD) Circuits, Civil Defence Blast Door (CDBD) Circuits, Next Train Out Indicator (NTOI) Circuits, Train Proceed Key Switch (TPKS) Circuits, ELD Circuits, WRU Circuits, PMI/MAU Rack Layouts, ATS/DCS Rack Layouts, Relay Rack Layout, Modular Rack Layout, Station Control Layout, Earth Distribution Layout, COC Layouts & Analysis, Cable schematics, Multicore Cable Details and Plug Coupler, CTF, DB Box, PFSE, Fuse, Contact & Terminal Analysis.
- Executed testing and commissioning of indoor and outdoor signaling gears for Stage 1 Nagole-Mettiguda.
- Performed indoor and outdoor testing of signaling gears at Uppal Depot.
- Conducted PICO (point-to-point continuity) testing of indoor and outdoor cable wiring, obtaining client approval.
- Tested Axle counters, signals, and point machines.
- Prepared and submitted test reports to the client.
- Generated scheme plans using the SGD (System Guide way & Data Generator) tool for BID support for THALES CANADA.
- Executed system performance simulation tests.

- Prepared Hyderabad Metro Application data.
- Conducted operational simulation testing of the Automatic Train using the TVS (Track & Vehicle Simulator) tool.
- Documented test reports and reviewed test logs, updating reports accordingly.
- Coordinated with consultants for the design approval of new signaling equipment.
- Supervised the installation of new signaling equipment at Riyadh Running Shed.
- Liaised with partners and subcontractors on behalf of the employer.
- Developed Testing & Commissioning (T&C) documents for Signaling and SCADA systems, obtaining approval from SRO (Saudi Railway Organization).
- Ensured proper installation and preventive maintenance of Rolling Stock.
- Oversaw outdoor installation work and participated in the testing and commissioning of outdoor machinery.
- Verified markings against SIP and discussed any non-conformities with the client.
- Prepared the necessary paperwork for work completion and obtained client authorization.
- Provided solutions to problems encountered by the site team.
- Traced and fixed Railway interlocking-related problems by utilizing maintenance tools and leveraging back-end knowledge of Application logic in the Microlok II.
- Installed various versions of maintenance tools on maintenance computers to interface with the MICROLOK II SSI System.
- Collected data dumps from the system, including system user data, event logs, and error logs.
- Conducted comprehensive analysis of data dumps and rectified any failures.
- Demonstrated complete understanding of the THALES SelTrac S40 CBTC system.
- Verified correct power input and start-up of an Axle Counter unit and STC.
- Adjusted an Axle Counter Unit EAK by connecting Hyperlink and adjusting MCON values.
- Verified block occupancy detection in Fallback Mode.
- Confirmed STC reaction to communication failures between ACE and Axle Counter Unit.
- Verified STC detection of the Platform Emergency Stop Device (PESB).
- Tested and commissioned the functioning of Platform Screen Doors units.
- Validated the station stopping accuracy of the train alignment with platform doors.
- Performed functional testing of automatic train movement.
- Prepared test reports and documentation in accordance with safety standards.
- Supervised contractor work for signaling gears installation.
- Managed cable laying, cable trenching, and marking of signaling gears.

Education

Bachelor of Engineering, Electrical and Electronics, OSMANIA UNIVERSITY, HYDERABAD, HYDERABAD

10/2004 – 05/2009

Skills

V-Cycle systems engineering

Requirements Management

Design Interface Compliance Management (EN50126, EN50128 and EN50129)

Interface Management

IVVQ

Installation

Principles Testing and Commissioning

System Design

System Safety Assurance

Training and Competency Management

Operations and Maintenance

End-to-End Project Lifecycle Management

Resident Engineer

Inspection & Test Planning (ITPs)

Risk Assessment

Design Review & Interface Coordination

Multicultural Team Leadership & Overseas Project Execution

Contractor Supervision

Overseas Project Delivery & Multicultural Team Collaboration

Client Collaboration & Interfacing

Stakeholder Engagement

Certificates

Professional Engineer, National Engineers Australia

Certified Trainer for ETCS Level 2 Signaling system, Thales Germany

Fatigue management and Fraud awareness trainings, Queensland Rail

Non-Electrician authorization training, APAVE

Track Access Safety awareness training, Thales Gulf